28-Day Top-Tier Entry-Level Cybersecurity Roadmap & Study Timetable

This comprehensive master schedule pairs the theoretical foundations of the **Google Cybersecurity Professional Certificate** with hands-on labs from **TryHackMe (THM)**. Designed specifically for aspiring SOC Analysts and Junior Cybersecurity Specialists, this blueprint transforms raw learning into job-ready technical capability and portfolio evidence within 4 weeks.

1. Master Schedule Overview

Phase / Week	Core Focus Area	Google Cert Mapping	TryHackMe Path & Modules	Core Job Skill Outcome
Week 1 (Days 1–7)	Foundations, Linux & Windows Administration	Courses 1 & 4	Pre-Security & Linux Fundamentals	OS Command Line Mastery, Security Frameworks (NIST, CIA)
Week 2 (Days 8–15)	Networking Protocols & Traffic Analysis	Course 3	Network Fundamentals, Nmap, Wireshark	Packet Capture Analysis, Network Reconnaissance & Port Scanning
Week 3 (Days 16–21)	Vulnerability Management & Python Security Scripting	Courses 5 & 7	OWASP Top 10, Cryptography, Python Basics	Web Application Vulnerabilities, Security Automation & Scripting
Week 4 (Days	SIEM, Incident	Course 6 & 8	Cyber Defense,	Log

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22–28)	Response & SOC Operations		Splunk, Snort, Forensics	Investigation, Threat Hunting, Incident Documentation

2. Daily Study Schedule Structure (4.5 Hours / Day)

To maximize retention and build muscle memory, structure each day into three distinct blocks:

- **Block 1: Theoretical Foundation (1.5 Hours)** — Complete video lectures, readings, and quizzes in the Google Cybersecurity Certificate on Coursera. Focus on core concepts, regulatory standards, and frameworks.
- **Block 2: Hands-on Practical Lab (2 Hours)** — Execute corresponding labs on TryHackMe inside the terminal/browser. Practice commands, solve challenges, and capture flags.
- **Block 3: Documentation & Portfolio Building (1 Hour)** — Document command outputs, write incident response notes, upload custom scripts to GitHub, or answer scenario questions.

3. Detailed Daily Execution Timetable

Day	Google Cert Module	TryHackMe Lab Room	Specific Skill & Deliverable
Day 1	Course 1: Foundations of Cybersecurity	Intro to Defensive Security	Understand CIA Triad, NIST CSF, security roles.
Day 2	Course 1: Security Domains & Ethics	Careers in Cyber	Identify core cybersecurity domains and ethics rules.
Day 3	Course 4: Linux Commands Basics	Linux Fundamentals Part 1	CLI Navigation: ls, cd, pwd, cat, find.

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Day 4	Course 4: Linux Permissions & Processes	Linux Fundamentals Part 2	File permissions (chmod, chown), process management.
Day 5	Course 4: Text Processing & Filtering	Linux Fundamentals Part 3	Piping, redirection, grep, cut, nano editor.
Day 6	Course 4: Windows Security Intro	Windows Fundamentals 1	Windows GUI/CLI, User Privileges, System Settings.
Day 7	Course 4: SQL Database Queries	SQL Fundamentals	Write basic SQL queries (SELECT, WHERE, JOIN).
Day 8	Course 3: Network Architecture	What is Networking?	Understand OSI Model layers vs. TCP/IP suite.
Day 9	Course 3: IP Addressing & Subnetting	Intro to Networking	IPv4/IPv6, CIDR notation, Subnet masks, MAC addresses.
Day 10	Course 3: Application Protocols	DNS in Detail & HTTP in Detail	Analyze DNS resolution queries and HTTP request/response headers.
Day 11	Course 3: Firewalls, VPNs & Routers	Network Services	Understand SMB, SSH, FTP, Telnet security configurations.
Day 12	Course 3: Passive Network Security	Passive Reconnaissance	Perform OSINT gathering using

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			WHOIS, nslookup, Dig.
Day 13	Course 3: Active Scanning Tools	Nmap Live Host Discovery	Execute ping sweeps and ARP scans to map active network hosts.
Day 14	Course 3: Port Scanning Techniques	Nmap Architecture & Flags	Master Nmap flags: -sS, -sT, -sV, -O, -p-.
Day 15	Course 3: Packet Analysis	Wireshark Basics	Inspect PCAP files, apply Wireshark display filters.
Day 16	Course 5: Asset Cryptography	Cryptography Basics	Symmetric vs. Asymmetric encryption, hashing (MD5, SHA256).
Day 17	Course 5: Threats & Vulnerabilities	OWASP Top 10 Intro	Explore top web vulnerabilities (Broken Access Control, Injection).
Day 18	Course 5: Web Application Security	SQL Injection & XSS	Identify SQLi paylods and Cross-Site Scripting attack vectors.
Day 19	Course 5: Password Security	Crack the Hash & Hydra	Use Hashcat/John the Ripper and Hydra for password audit tests.
Day 20	Course 7: Python Data Types & Control	Python Basics	Write scripts using variables, loops, conditional statements.

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			Certificate.

4. Portfolio Projects for Resume & Top-Tier Readiness

To stand out to recruiters, build and publish these 3 essential portfolio projects on GitHub during your 28-day study period:

1. **Automated Log Parser (Python):** A Python script that reads Linux /var/log/auth.log or Web server access logs, identifies failed login attempts, counts occurrences per IP, and flags IPs exceeding threshold limits.
2. **Network Reconnaissance Report (Nmap & Wireshark):** A detailed PDF report documenting a vulnerability assessment of a sandbox environment on THM, featuring port scan analysis, service version enumeration, and severity scoring.
3. **SOC Incident Report & Playbook:** A professionally formatted SOC triage document covering a simulated ransomware attack or breach on Splunk, including timeline analysis, compromised accounts, indicators of compromise (IOCs), and remediation recommendations.

5. Top-Tier Technical Interview Checklist

Be prepared to answer these core technical interview questions by Day 28:

- Explain the 3-way handshake in TCP/IP (SYN, SYN-ACK, ACK).
- What is the difference between a stateful and stateless firewall?
- Walk me through how you would investigate a suspicious alert showing 500 failed SSH logins in 1 minute.
- Explain Cross-Site Scripting (XSS) vs. SQL Injection (SQLi) and how to mitigate both.
- How do you use Splunk SPL to filter traffic by a specific IP address and timestamp?